RSCL — Algorithm 1: HU Risk-Asymmetric Confidence Gating and Command-Intent Mismatch Checks

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Date: 2026-11-12 · Document type: formal algorithm specification of the implemented code (HU-MA'AT drone-fleet simulation demo)

Source files described (content hash and state at document build time):

`rscl-hu-latest/hu/gateway/hu_service.py`	`sha256 a672156992ef`	2026-11-11 01:41 UTC	clean @ 091441f
`hu-telepresence-gazebo/festival/hu_live_server.py`	`sha256 2416370e93dc`	2026-11-11 06:38 UTC	clean @ 10ee643
`hu-telepresence-gazebo/festival/control_server_hu.py`	`sha256 3f32578a144c`	2026-11-11 06:32 UTC	clean @ 10ee643
`hu-telepresence-gazebo/festival/mission_intents.py`	`sha256 e109c5a9e89c`	2026-11-12 01:11 UTC	clean @ 05172be
`hu-telepresence-gazebo/festival/hu_live_dashboard.html`	`sha256 d1e2cf6aabae`	2026-11-12 01:23 UTC	clean @ 05172be

1. Purpose and context

HU is the natural-language command interpreter of the HU-MA'AT demo. A frozen pretrained language model (SmolLM2-135M) supplies sentence representations; three small trained heads classify the flight action, decide per aircraft whether it is addressed, and detect compound commands. This algorithm turns those neural outputs into one of three decision bands, **EXECUTE**, **CONFIRM** (operator must confirm) or **REFUSE**, and then applies deterministic guards that can only *lower* a band (never raise it). The asymmetry is deliberate: a false refusal costs the operator a repeat; a false execution moves an aircraft. The same thresholds are re-checked independently at the flight supervisor's HTTP boundary.

The model path is reached only after the symbolic emergency channel (Algorithm 2) and the deterministic mission router (Algorithm 3) have declined the utterance. In the current routing, the sentences that reach the neural path and can be dispatched are, in practice, plain take-off / land / hover / stop primitives addressed to names, collectives or exclusions; moves, turns and formations are handled by the mission route.

2. Inputs, outputs and data structures

Item	Definition
Input u	Utterance string, 1..300 characters (typed, or voice after review / clarification, see Algorithm 19).
Roster R	Ordered names from `festival/hu_sim_roster.yaml` (assignments): Horus, Ra, Thoth, Anubis, Osiris. If the file has no assignments, `HU.roster()` falls back to [Horus, Ra, Thoth].
Action set A	TAKEOFF, LAND, HOVER, MOVE, TURN, UP, DOWN, STOP, REFUSE (9 classes).
Parse record	`{action, params, targets, per_drone, confidence, band, reason, latency_ms}` from `HU.parse`.
Pending table	`Service.pending[cmd_id] = {action, targets, params, confidence, at}`; single-use, expires after 90 s.
Output	A logged decision record (append-only `logs/hu_real_decisions.jsonl`) with band, reason, `dispatched` flag; for EXECUTE an HTTP dispatch to `/api/hu/dispatch`.

3. Algorithm

Algorithm 1.1 HU-Parse (neural band assignment)

Source: rscl-hu-latest/hu/gateway/hu_service.py: HU.parse, HU._emb, _mentions, extract_params

```
procedure HU-PARSE(u)
  R ← roster()                                        // ordered aircraft names
  T ← [u] ++ [ "drone: r | index: k (ordinal words) | mentioned: M(r,u) | utterance: u" for k, r in R ]
  E ← EMB(T)                                          // frozen SmolLM2-135M, truncation 64 tokens
                                                      // EMB = concat(masked mean, masked max) of last hidden state
  p_a ← softmax(H_action(E[0]));  a ← argmax p_a;  c_a ← max p_a
  comp ← sigmoid(H_compound(E[0])) > 0.5
  for i in 1..|R| do  m_i ← sigmoid(H_match(E[i]))
  targets ← [ R_i : m_i > τ ]                         // τ from checkpoint (key "tau"), default 0.5
  c_t ← min_i max(m_i, 1 − m_i)                       // weakest per-aircraft decision (1.0 if no aircraft)
  c ← min(c_a, c_t)
  if comp then band ← REFUSE                          // "compound commands: coming online"
  else if a = REFUSE then band ← REFUSE
  else if targets = ∅ then band ← REFUSE              // no addressee resolved
  else if c ≥ 0.85 then band ← EXECUTE
  else if c ≥ 0.55 then band ← CONFIRM
  else band ← REFUSE
  params ← EXTRACT-PARAMS(a, u)                       // symbolic; may set params.range_violation
  return (a, params, targets, c, band)
```

`M(r,u)` (`_mentions`) is "yes" if the lower-cased name is a substring of *u* or some token differs from it by at most one character (position-wise mismatches plus length difference, lengths within one), else "no". `EXTRACT-PARAMS` normalises number words and units (m → cm, ft → cm, inch → cm, min → s), takes the first integer, gates it to a safe range, and snaps it to a bin; an out-of-range value sets `range_violation` and leaves the parameter empty.

Algorithm 1.2 Bridge guards and band enforcement (neural route only)

Source: festival/hu_live_server.py: Service.command (model branch), explicit_primitive_intent, stated_targets, translate, Service.confirm

```
procedure HU-GUARD(u_norm, unrecognized)       // reached only if Alg. 2 and Alg. 3 declined u
  r ← HU-PARSE(u_norm)
  I ← EXPLICIT-PRIMITIVE-INTENT(u_norm)          // lexical candidate set or ∅
  if unrecognized and r.band ≠ REFUSE then r.band ← REFUSE   // no explicit flight command: never execute
  if I ≠ ∅ and r.action ∉ I then r.band ← REFUSE             // command-intent mismatch
  X ← STATED-TARGETS(u_norm)                     // None if no explicit address
  if X ≠ None and set(r.targets) ≠ set(X) then r.band ← REFUSE   // named-address guard
  if r.params.range_violation then r.band ← REFUSE
  if r.band ∈ {EXECUTE, CONFIRM} then
    try
      (act, pr) ← TRANSLATE(r.action, r.params)  // raises on any unsupported / out-of-range value
      require r.targets non-empty and r.targets ⊆ NAMES
      require 0 ≤ r.confidence ≤ 1
      if r.band = EXECUTE then
        require r.confidence ≥ 0.85; dispatch(act, r.targets, pr, r.confidence, confirmed=false)
      else
        require r.confidence ≥ 0.55; pending[id] ← (act, r.targets, pr, r.confidence, now)
    except ValueError or DispatchError: r.band ← REFUSE
  log r; return r
procedure CONFIRM(id)                            // operator action
  p ← pending.pop(id)                            // single use
  if p = None or now − p.at > 90 s then refuse
  dispatch(p.act, p.targets, p.params, p.confidence, confirmed=true)
```

Algorithm 1.3 Supervisor-side gate

Source: festival/control_server_hu.py: Handler.do_POST, route /api/hu/dispatch

```
procedure HU-DISPATCH(request)
  if not constant_time_equal(request.X-HU-Token, token_file) then return 403
  c ← float(request.confidence); conf_ok ← finite(c) and 0 ≤ c ≤ 1
  if not conf_ok then return 400
  emergency ← request.symbolic_emergency = true and action ∈ {emergency_land, emergency_hover}
  if not emergency and (c < 0.55 or (c < 0.85 and not request.operator_confirmed)) then
    return 409 (decision CONFIRM if c ≥ 0.55 else REFUSE)
  return Fleet.command(action, targets, params)  // all supervisor preconditions still apply
```

Lexical guards

- **EXPLICIT-PRIMITIVE-INTENT**: adds TAKEOFF for *take off / launch / lift off*; LAND for *land / landing / touch down*; {STOP, HOVER} for *stop / hover / hold*; MOVE for *forward, back(ward)(s), move, left, right*; UP for *go up, rise, ascend, up, climb*; DOWN for *go down, descend, down, lower*. Multiple words give a union; the guard only fails when the predicted action is outside the union.
- **STATED-TARGETS**: exact whole-word fleet names; a collective (*everyone, everybody, all (the) drone(s)*, or *all* followed by a primitive verb) gives the whole fleet minus one recognised exclusion (*except / excluding / but not* + name). An exclusion word without a recognised name returns None (the guard abstains and the model band stands).
- **TRANSLATE**: TAKEOFF → takeoff at 1.65 m; LAND → land; STOP and HOVER → hover; MOVE distance must be 10..200 cm, direction forward/back/left/right; UP/DOWN 10..200 cm; TURN 5..180° clockwise/counterclockwise; any other label is refused.

4. Flowchart and state diagram

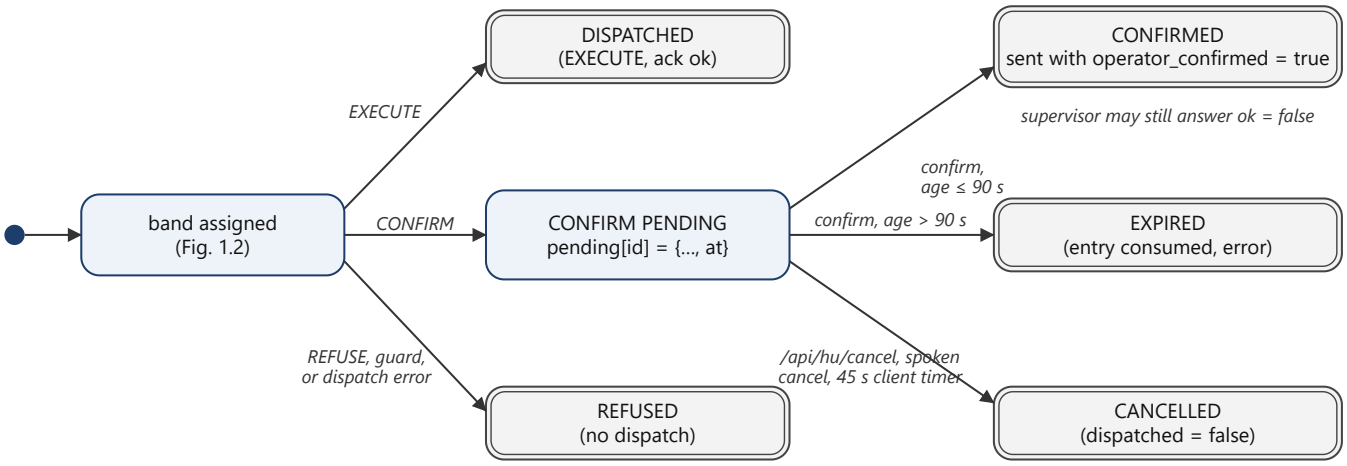


Figure 1.1. State machine of one decision (`Service.pending`, `Service.confirm`, `Service.cancel`). Entries are single use: `pending.pop` runs before the age test, so a late or repeated confirm finds nothing and is refused. There is no server-side sweeper; an entry that is never confirmed or cancelled stays in memory until the process restarts (it can no longer be dispatched after 90 s). The 45 s timer is the hands-free dashboard's own timeout, which sends `/api/hu/cancel`. Mission proposals (Algorithm 3) use the same states with `kind = mission` and are sent to `/api/mission/dispatch`.

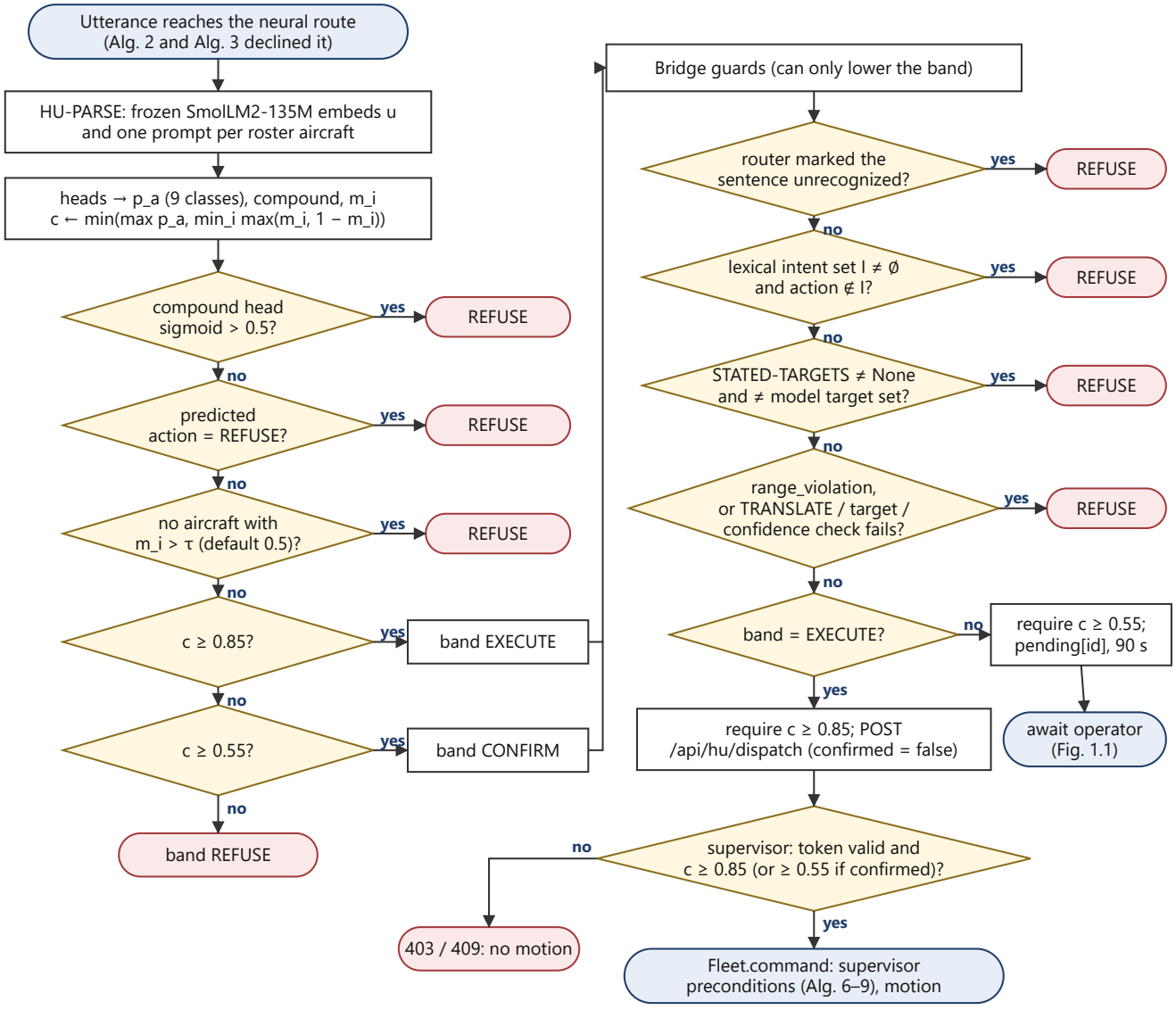


Figure 1.2. Flowchart of the neural route (Algorithms 1.1–1.3). Every REFUSE box is a logged decision record with `dispatched = false`. Thresholds 0.85 / 0.55 are applied three times: in `HU.parse`, in the bridge, and at `/api/hu/dispatch`.

5. Parameters

Parameter	Value	Where
Backbone	HuggingFaceTB/SmolLM2-135M, frozen, eval mode, max 64 tokens	hu_service.py MODEL
Head architecture	Linear(d,384) – GELU – Dropout 0.1 – Linear(384,k); k = 9 / 1 / 1	hu_service.Head
Checkpoint	`hu/models/hu_heads_v1_1.pt`, else `hu_heads_v1.pt`	hu_live_server.main
EXECUTE threshold	0.85 (model, bridge and supervisor)	HU band_exec; Service.command; /api/hu/dispatch
CONFIRM threshold	0.55 (model, bridge and supervisor)	HU band_confirm; same
Match threshold τ	checkpoint value, default 0.5	HU.__init__

Compound threshold	0.5 on sigmoid	HU.parse
Parameter gates / bins	MOVE, UP, DOWN 10..200 cm, bins {20,30,50,75,100,150}, defaults 50 / 30 cm; TURN 5..360°, bins {15,45,90,135,180}, default 90; HOVER 1..15 s, bins {2,3,5,10}, default 3	extract_params
Bridge translation bounds	moves 10..200 cm; turns 5..180°; take-off altitude 1.65 m	translate
Confirmation lifetime	90 s, single use	Service.confirm
Utterance length	1..300 characters; HTTP body ≤ 4096 bytes	Service.command; Handler.do_POST

6. Complexity

One batched encoder forward pass over |R|+1 sequences of at most 64 tokens: $O((|R|+1)\cdot L^2\cdot d)$ attention cost with $L \le 64$, plus O(|R|) head evaluations. All guards are regular-expression scans, O(|u|) each. Measured per-parse latency is reported in each decision record (`latency_ms`); no aggregate is computed here.

7. Properties

Guaranteed by the code

- **Monotone refusal.** Every guard after `HU.parse` can only set the band to REFUSE; none can promote REFUSE or CONFIRM.
- **No dispatch below 0.85 without confirmation, and none below 0.55 at all** (outside the symbolic emergency channel): enforced in the bridge and again, independently, at the supervisor (constant-time token check, finite confidence in [0,1]).
- **Joint confidence.** The reported confidence is the minimum of the action probability and the weakest per-aircraft address decision, so an uncertain address lowers the band even when the action is certain.
- **Contradiction refusal.** If the text explicitly names a primitive verb family, a prediction outside that family is never dispatched, whatever its confidence. If the text explicitly names aircraft or a collective, any different target set is refused rather than repaired.
- **Unrecognised sentences never execute**: when the mission router marks a sentence as having no explicit flight command, a non-REFUSE model output is overridden to REFUSE.
- Confirmations are single-use and expire after 90 s.

Not guaranteed

- The thresholds are applied to softmax/sigmoid outputs; no calibration of these scores is performed or verified in this repository. "Confidence" is therefore a score, not a probability of correctness.
- When the utterance contains no explicit verb or address, the lexical guards abstain, and the decision rests on the model alone (e.g. "up you go" relies on the trained heads).
- The TURN class has no lexical intent family; a TURN prediction is constrained only by TRANSLATE bounds. (Turns are normally routed to the mission parser first.)
- The HOVER `duration` parameter is extracted but not used by TRANSLATE; hover is an indefinite hold.
- **Exclusion lists in the address guard.** STATED-TARGETS recognises only the first excluded name, and treats plain *all* as a collective only when a primitive verb follows it directly. For "All except Ra and Thoth, take off" (a plain primitive, so it is routed to HU by Algorithm 3) it returns the named set {Ra, Thoth}, i.e. the aircraft the operator excluded. A correct model target set {Horus, Anubis, Osiris} is then refused (safe direction), but a

model error that selects {Ra, Thoth} would pass this guard. Found by calling the function offline; not covered by a test (see Algorithm 4, Table 4.1).
- Physical safety of the dispatched action is not decided here; it is delegated to the supervisor's preconditions (Algorithms 6–9).
- Roster fallback [Horus, Ra, Thoth] would silently shrink the fleet if the roster file lost its assignments.

8. Testing and measured results

- Offline (no Gazebo): `acceptance/tests/test_v9.py` (e.g. `test_wrong_model_label_refused_even_high_confidence`, `test_real_hu_still_executes_matching_label`, `test_unrecognized_sentence_shows_hu_but_never_executes`, `test_original_primitive_direct_hu_route`) with a mock HU; `acceptance/tests/test_http_v9.py` (`test_hu_threshold_unchanged`, `test_unauthorized_hu_refused`). Run by `acceptance/run_offline.sh`; the most recent recorded run (`acceptance/results/run_20261111T052914Z/offline.txt`) reports 44 tests, OK.
- Live, non-actuating: `acceptance/command_matrix.py`, 28/28 PASS in the same run. Note: that run predates the latest edit of `mission_intents.py` (plain-primitive routing to HU), so it measured the previous routing.
- Live diagnostics: `acceptance/hu_band_probe.py`, `acceptance/hu_intent_probe.py`.
- Accuracy, calibration and threshold sensitivity of the trained heads: **not measured here**.

9. Comparison with alternatives and related work

Turning a classifier score into execute / ask / refuse is the reject-option or selective-prediction problem. Four established approaches are compared with the gate of this document.

- **Chow's reject rule** [1]: with true class posteriors, rejecting whenever the top posterior is below a threshold derived from the costs of error and rejection minimises expected cost. HU's two thresholds are a three-way extension of the same form, but HU's scores are not posteriors (no calibration is performed), so the optimality argument does not transfer.
- **Selective classification with a risk guarantee** [2]: the threshold on the maximum softmax response is chosen on held-out labelled data so that the selective error is below a user-set target with stated probability. HU's 0.85 / 0.55 are fixed constants; no risk–coverage curve has been measured for them.
- **Post-hoc calibration** (temperature scaling) [3]: one scalar fitted on a validation set makes softmax confidences approximately calibrated. It is not applied to HU's action or match heads.
- **Conformal prediction** [4], and its use for language-driven robots in KnowNo [5]: a calibration set yields prediction sets with a distribution-free coverage guarantee (under exchangeability); the robot asks a human whenever the set is not a single option. HU's CONFIRM band plays the same role (ask the operator), without the coverage guarantee.

Approach	Decision rule	Formal guarantee	Needs	HU gate, by comparison
Chow reject rule [1]	reject if max posterior < t(costs)	minimum expected cost, if posteriors are exact	calibrated posteriors, cost model	same shape, two thresholds; scores uncalibrated, costs not modelled
Selective classification, SR [2]	threshold on max softmax, fitted to target risk	selective risk ≤ r* w.p. 1−δ (i.i.d. data)	labelled held-out set	hand-set thresholds; no risk measured
Temperature scaling [3]	rescale logits before thresholding	none; empirical calibration	validation set	not applied
Conformal sets / KnowNo [4, 5]	ask for help unless the set is a singleton	marginal coverage ≥ 1−α under exchangeability	calibration set	CONFIRM band asks; no coverage statement

This document	min(action, address) score vs 0.85 / 0.55, then deterministic guards that can only refuse	code-level: monotone refusal, triple threshold check	nothing beyond the trained heads	—

Where the HU gate is weaker. None of its thresholds has a statistical interpretation, and the error rate of EXECUTE decisions is unknown; [2]–[5] would each require a labelled, representative set of operator utterances, which does not exist in this repository. **Where it differs.** The lexical intent check, the exact-address guard and the supervisor re-check are independent of the score, so a confidently wrong action label or target set is refused when the sentence itself contradicts it; the score-based methods above do not include such checks. The guards do not help when the sentence contains no explicit verb or address.

10. Relation to other algorithms

Preceded by Algorithm 2 (emergency channel, which bypasses this gate) and Algorithm 3 (routing). Uses address semantics close to, but not identical with, Algorithm 4. Dispatched commands are subject to Algorithms 6–9 in the supervisor. Voice input reaches this gate only through Algorithm 19.

11. References

1. C. K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Trans. Information Theory*, 16(1):41–46, 1970. `https://doi.org/10.1109/TIT.1970.1054406`
2. Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," *NeurIPS* 2017, pp. 4878–4887. `https://arxiv.org/abs/1705.08500`
3. C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, "On Calibration of Modern Neural Networks," *ICML* 2017, PMLR 70:1321–1330. `https://proceedings.mlr.press/v70/guo17a`
4. A. N. Angelopoulos and S. Bates, "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification," arXiv:2107.07511, 2021. `https://arxiv.org/abs/2107.07511`
5. A. Z. Ren, A. Dixit, A. Bodrova, et al., A. Majumdar, "Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners," *CoRL* 2023, PMLR 229:661–682. `https://proceedings.mlr.press/v229/ren23a.html`

This specification describes the behaviour of the source revision identified above. Line-level references use `file:function`. Where the code and its own comments disagree, the code is described. Simulation-only system: no statement here is a claim about flight hardware.